



Epidemiology and Key Points

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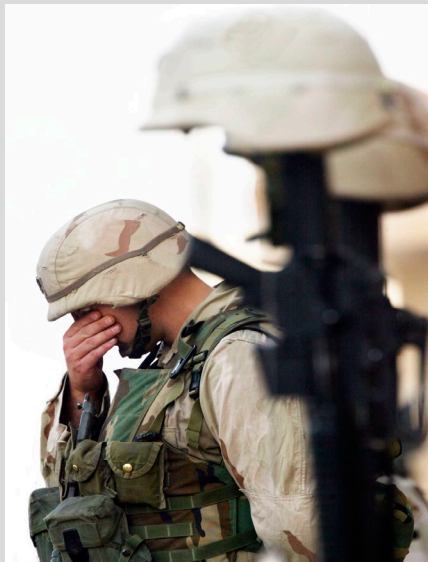
- Likelihood of Experiencing 1 traumatic event over lifespan
 - 60% of men
 - Most often accidents, physical assault, combat, disaster, or witnessing death or injury
 - 50% of women
 - Most often sexual assault and child sexual abuse
- 7-8% of US population will develop PTSD at some point in their lives
- An estimated 3.9% of the world population has had post-traumatic stress disorder(PTSD) at some stage in their lives
- Most people exposed to potentially traumatic events do not develop PTSD
- Feeling supported by family, friends or other people following the potentially traumatic event can reduce the risk of developing PTSD.
- More women are affected by PTSD than man.

What is Trauma?

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- “An emotional response to a terrible event”
 - These events include military violence, sexual violence, disasters, and accidents





Acute Stress Reaction

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- Acute Stress Reaction
- Within **1 month** of a traumatic event
 - Intrusion, negative mood, dissociation, avoidance, and arousal
- Long Term
 - Unpredictable emotions, flashbacks, strained relationships, physical symptoms

<http://www.apa.org/topics/trauma/>

Depress Anxiety. 2011 Sep;28(9):802-17. doi: 10.1002/da.20737. Epub 2010 Nov 3

A 32 year old Iraq war veteran reports to your primary care office. He began his military career at 18 years old, with a deployment to Afghanistan in 2003. Since then he has done four tours in Iraq and Afghanistan. He has been involved in **multiple raids** and **firefights**. He recently retired from the military and has had a hard time returning to civilian life.



Example Case – Criterion A



Criterion A: Stressor

The person was exposed to: death, threatened death, actual or threatened serious injury, or actual or threatened sexual violence, as follows: (one required)

1. Direct exposure.
2. Witnessing, in person.
3. Indirectly, by learning that a close relative or close friend was exposed to trauma. If the event involved actual or threatened death, it must have been violent or accidental.
4. Repeated or extreme indirect exposure to aversive details of the event(s), usually in the course of professional duties (e.g., first responders, collecting body parts; professionals repeatedly exposed to details of child abuse). This does not include indirect non-professional exposure through electronic media, television, movies, or pictures.

While questioning your patient he begins to describe
awakening frequently from nightmares, as well as
feeling like he is “back in the field, randomly, even
when I am just hanging out with friends at a bar”

Case Continued – Criterion B



Criterion B: Intrusive Symptoms

The traumatic event is persistently re-experienced in the following way(s): **(one required)**

1. Recurrent, involuntary, and intrusive memories. Note: Children older than six may express this symptom in repetitive play.
2. Traumatic nightmares. Note: Children may have frightening dreams without content related to the trauma(s).
3. Dissociative reactions (e.g., flashbacks) which may occur on a continuum from brief episodes to complete loss of consciousness. Note: Children may reenact the event in play.
4. Intense or prolonged distress after exposure to traumatic reminders.
5. Marked physiologic reactivity after exposure to trauma-related stimuli.

The patient has also found that he no longer is comfortable seeing his friends from his troop nor going out to crowded places. He spends most of his time at home. When you ask him about this, he says he sometimes feels like he is back in Iraq when his in a large crowd.

Case Continued – Criterion C



Criterion C: Avoidance

Persistent effortful avoidance of distressing trauma-related stimuli after the event: **(one required)**

1. Trauma-related thoughts or feelings.
2. Trauma-related external reminders (e.g., people, places, conversations, activities, objects, or situations).

You question the patient about the details of the raid that he describes keeping him up at night. He says he cannot remember about five minutes in the middle of the event. He is disturbed by the fact that he cannot accurately recall what happened during that time period. As you continue to talk, he says he find little pleasure in the hobbies he used to enjoy, like playing basketball and seeing movies with friends.

Case Continued – Criterion D



Criterion D: Negative Alterations in Cognitions and Mood

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Negative alterations in cognitions and mood that began or worsened after the traumatic event: **(two required)**

1. Inability to recall key features of the traumatic event (usually dissociative amnesia; not due to head injury, alcohol, or drugs).
2. Persistent (and often distorted) negative beliefs and expectations about oneself or the world (e.g., "I am bad," "The world is completely dangerous").
3. Persistent distorted blame of self or others for causing the traumatic event or for resulting consequences.
4. Persistent negative trauma-related emotions (e.g., fear, horror, anger, guilt, or shame).
5. Markedly diminished interest in (pre-traumatic) significant activities.
6. Feeling alienated from others (e.g., detachment or estrangement).
7. Constricted affect: persistent inability to experience positive emotions.

Source: **DSM-5 APA (2013); PTSD: National Center for PTSD**

Another reason your patient finds it difficult to be in large crowds, is that he's “**constantly scanning around**” to keep an eye on everyone there. He feels unsure about his safety and the safety of those around him. He describes a particularly distressing day when **he heard a car back fire, which made him jump up and take off running down the block. It felt like his heart was beating out of his chest.**

Case Continued – Criterion E



Criterion E: Alterations in Arousal and Reactivity

Trauma-related alterations in arousal and reactivity that began or worsened after the traumatic event: **(two required)**

1. Irritable or aggressive behavior
2. Self-destructive or reckless behavior
3. Hypervigilance
4. Exaggerated startle response
5. Problems in concentration
6. Sleep disturbance

When asked about how long he has been feeling this way your patient says all of these symptoms have been present for the last **six months** and seem to be getting worse, not better.

Case Continued – Criterion F

Criterion F: Duration

Persistence of symptoms (in Criteria B, C, D, and E)
for more than **one month**.

The patient has come to you because he knows he needs to find a job, but he **cannot seem to find it in himself to go out for interviews**. He's also become **very isolated** from his friends and family.

Case Continued – Criterion G

Criterion G: Functional Significance

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Significant symptom-related distress or functional impairment (e.g., social, occupational).

Source: **DSM-5 APA (2013); PTSD: National Center for PTSD**

He has no significant past medical history, he is not on any medications, he had lab work done when he returned home and was told he is “fit as can be”. He rarely drinks, denies recreational drug use, and smokes 10 cigarettes per day.

Case Continued - Criterion H

Criterion H: Exclusion

Disturbance is not due to medication, substance use, or other illness.

With the information you have gathered, you decide this patient may have PTSD and you need to assess further.

Case Continued – Further assessment

DSM-V PTSD Subtypes



Specify if: With Dissociative Symptoms.

In addition to meeting criteria for diagnosis, an individual experiences high levels of either of the following in reaction to trauma-related stimuli:

- 1. Depersonalization:** experience of being an outside observer of or detached from oneself (e.g., feeling as if "this is not happening to me").
- 2. Derealization:** experience of unreality, distance, or distortion (e.g., "things are not real" "this is more like a dream").



Specify if: With Delayed Expression.

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Full diagnosis is not met until at least six months after the trauma(s), although onset of symptoms may occur immediately.

Source: **DSM-5 APA (2013); PTSD: National Center for PTSD**



Specify if: **Preschool Subtype**

- Children 6 years or younger
 - Immediate reaction is not assessed. Children may not be able to express that they've been exposed to a traumatic event, if there were no adults around.
- Intrusive symptoms
 - Research has shown children do not always manifest with overt intrusive symptoms
- Avoidance and Negative alterations
 - Highly internalized phenomena
 - Only 1 symptom is required (instead of 3)
- Increased Arousal Symptoms
 - Most easily recognizable – extreme tantrums



Assessment Tools: Primary Care PTSD Screen (PC-PTSD)

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- A series of Yes/No Questions, Considered a **POSITIVE** screen if 3 or more answers are yes.

In your life, have you ever had any experience that was so frightening, horrible, or upsetting that, in the past month, you:

1. Have had nightmares about it or thought about it when you did not want to?
 2. Tried hard not to think about it or went out of your way to avoid situations that reminded you of it?
 3. Were constantly on guard, watchful, or easily startled?
 4. Felt numb or detached from others, activities, or your surroundings?
- A positive test DOES NOT confirm PTSD, but rather that further investigation is necessary.

Assessment Tools: PC-PTSD

Clarification

If the PC-PTSD screening instrument is utilized, clarify responses to determine:

- **Whether the patient has had a traumatic experience.**

"I notice from your answers to our questionnaire that you experience some symptoms of stress. At some point in their lives, many people have experienced extremely distressing events such as combat, physical or sexual assault, or a bad accident, and sometimes those events lead to the kinds of symptoms you have. Have you ever had any experiences like that?"

Assessment Tools: PC-PTSD

Clarification

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- **Whether reported symptoms are really trauma-related.**

"I see that you have said you have nightmares about or have thought about an upsetting experience when you did not want to. Can you give me an example of a nightmare or thinking about an upsetting experience when you didn't want to?" If a patient gives an example of a symptom that does not appear to be in response to a traumatic event (e.g., a response to a divorce rather than to a traumatic event), it may be that he or she is ruminating about a negative life event rather than experiencing intrusive thoughts about a traumatic stressor.

- **Whether reported symptoms are disruptive to the patient's life.**

"How have these thoughts, memories, or feelings affected your life? Have they interfered with your relationships? Your work? How about with recreation or your enjoyment of activities?"

Treatments

- Cognitive Behavioral Therapy (CBT)
 - Most well studied form of treatment
 - Helps to correct faulty cognitions, while attenuating symptoms, through behavioral exposure.
 - Treatment often includes cognitive therapy, exposure, and coping skills training.
- Examples of Cognitive Behavioral Therapies
 - Prolonged Exposure (PE)
 - Cognitive Processing Therapy (CPT)
 - Eye Movement Desensitization and Reprocessing (EMDR)



Treatment - Prolong Exposure Therapy

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- Prolonged Exposure therapy
 - Survivors repeatedly reexperience traumatic events via imaginal and in vivo exposure in safe situations
 - Trauma can become reprocessed and less painful
 - Following imaginal exposure, the patient will process with clinician to establish the emotional response to the experience in this safe environment
 - Patients are educated about common reactions to trauma
 - Usually administered for 9-12 weeks in twice weekly 90 Min Sessions
- Efficacy - More than 23 Randomized Control Trials have shown clinically significant improvements



Treatment - Cognitive Processing Therapy

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- Cognitive Processing Therapy
 - Principally a cognitive therapy
 - Exposure is achieved through written accounts of the traumatic event
 - Investigates the feelings elicited by the writing experience
 - Trains patient to challenge Faulty beliefs
- Efficacy – CPT vs PE or control showed as effective in reducing symptoms



Treatment - EMDR

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- Eye Movement Desensitization Reprocessing
 - CBT that incorporates saccadic eye movements
 - The patient imagine a traumatic event, while focusing on their feelings and cognitions
 - The clinician moves two fingers in front of their eyes to induce saccadic eye movements, until anxiety abates, then the patient is instructed to produce a more adaptive thought about the Event.
- Efficacy – remains uncertain, but some organizations recommend it as a treatment option.



Treatments - Pharmacotherapies

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- For people with treatment resistant PTSD, especially, combinations of Therapy and Medications are indicated
- SSRIs (first-line)
 - Paroxetine and Sertraline
 - Some evidence that more resilient patients will respond quickly
- Other Options
 - SSRI Fluoxetine
 - SNRI (Venlafaxine)
 - Other Antidepressants (Mirtazapine, Trazadone)
 - Prazosin (Nightmares)



Alternative Treatments

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- **Complementary and Alternative Medicine (CAM)**
 - Widely used, outside the standard practice
 - Includes Yoga, Acupuncture, meditation, Tai Chi, music, art, dance, and equine therapy
 - Estimates of 25-50% of patients are implementing
 - Current studies underway, but no solid evidence currently
- **Mindfulness Based Stress Reduction (MBSR)**
 - 9 weeks, group practice sessions
 - Teaches attending to the present moment in a non-judgmental, accepting way
 - Recent study showed improved subjective experience



Prognosis/Course

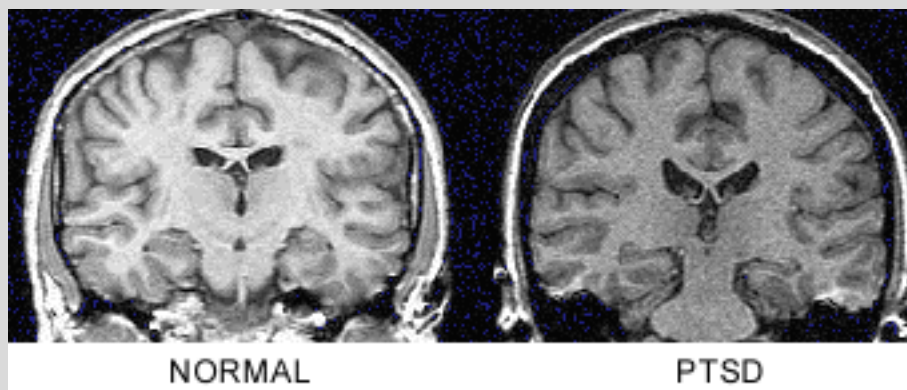
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- PTSD is often Chronic
 - 1/3 of patients recover in one year
 - 1/3 are symptomatic after ten years
- Established Relationship between Physical Health and Trauma
 - Evidence that development of PTSD may be the link
- Theories that neurochemical changes occur in PTSD
 - May have biological effect, for example increased HTN and Atherosclerotic Disease
- Mortality – Follow-up to NVVRS
 - From initial to follow-up 30 years later, Vietnam Veterans with PTSD were 2x as likely to have died.

Research - Imaging

- Imaging studies have shown:
 - Decreased volume, neuronal integrity, and functional integrity of the hippocampus
 - Increased amygdala activity – positively correlated with Severity of PTSD
 - Inverse relationship with Medial Prefrontal cortex activity



Source: Shin, L.M., Rauch, S.L. and Pitman, R.K., 2006. Amygdala, medial prefrontal cortex, and hippocampal function in PTSD. *Annals of the New York Academy of Sciences*, 1071(1), pp.67-79.



Research – Neurobiology and Genetics

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- Studied as Biomarkers for Disease
- Many neurobiological changes have been observed in PTSD research
 - Among these: HPA axis dysregulation, enhanced glutamatergic release, decreased serotonergic release, thyroid function disturbance, increased sympathetic response (hyperarousal/hypervigilance)



Research and Novel Treatment

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- Evidence-based trials
 - Three elements: Best evidence-based research, clinical judgment, and patient preference
 - Studies have shown PE and CPT – the gold standard of treatment are not useful for all patients
- New studies looking at Transcranial Magnetic stimulation, Virtual Reality Exposure Therapy, Deep Brain Stimulation
- MDMA, Ketamine, Psilocybin, CBD with assisted exposure/psychotherapy

Source: Steenkamp, MM. True evidence based care for Posttraumatic Stress Disorder in the military. JAMA Psychiatry. 73(5):431-432, May 2016.

Summary

- A majority of people will experience trauma over their lifetime.
- There can be serious social, emotional, occupational, and health implications to the development of PTSD.
- Key to understanding PTSD is identifying patients through screening and treating them appropriately.
- A wide body of research exists and is ongoing. Key to the future is identifying what makes a person vulnerable or resilient.

Lecture Feedback Form:

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<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>